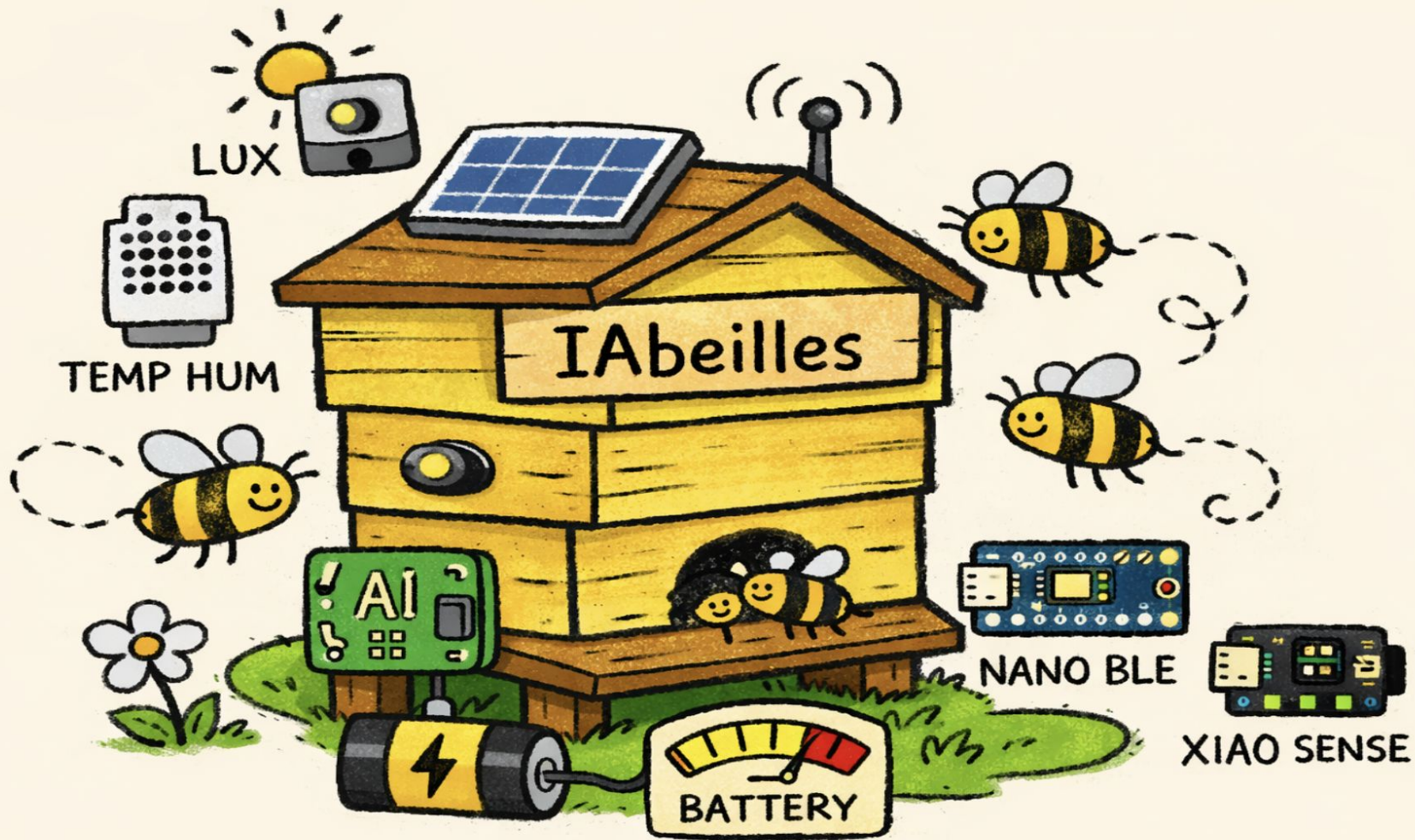
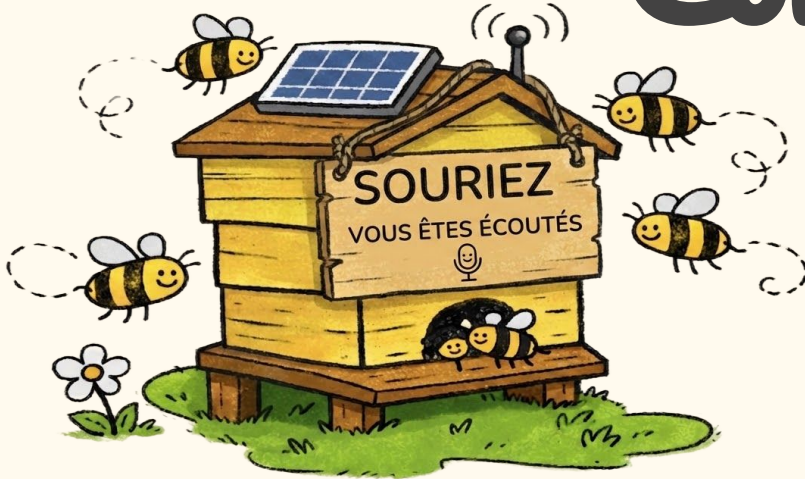
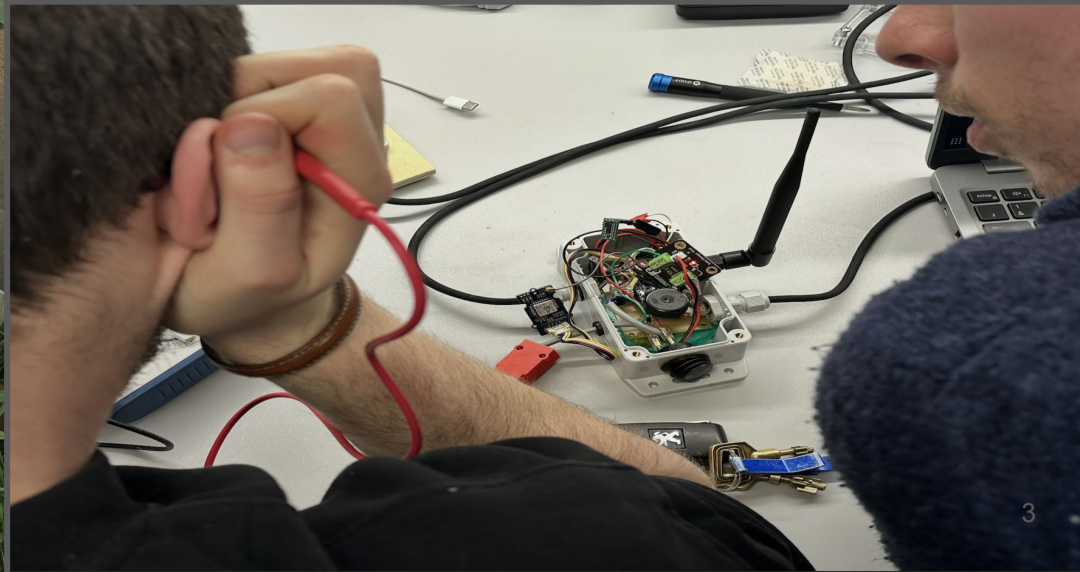




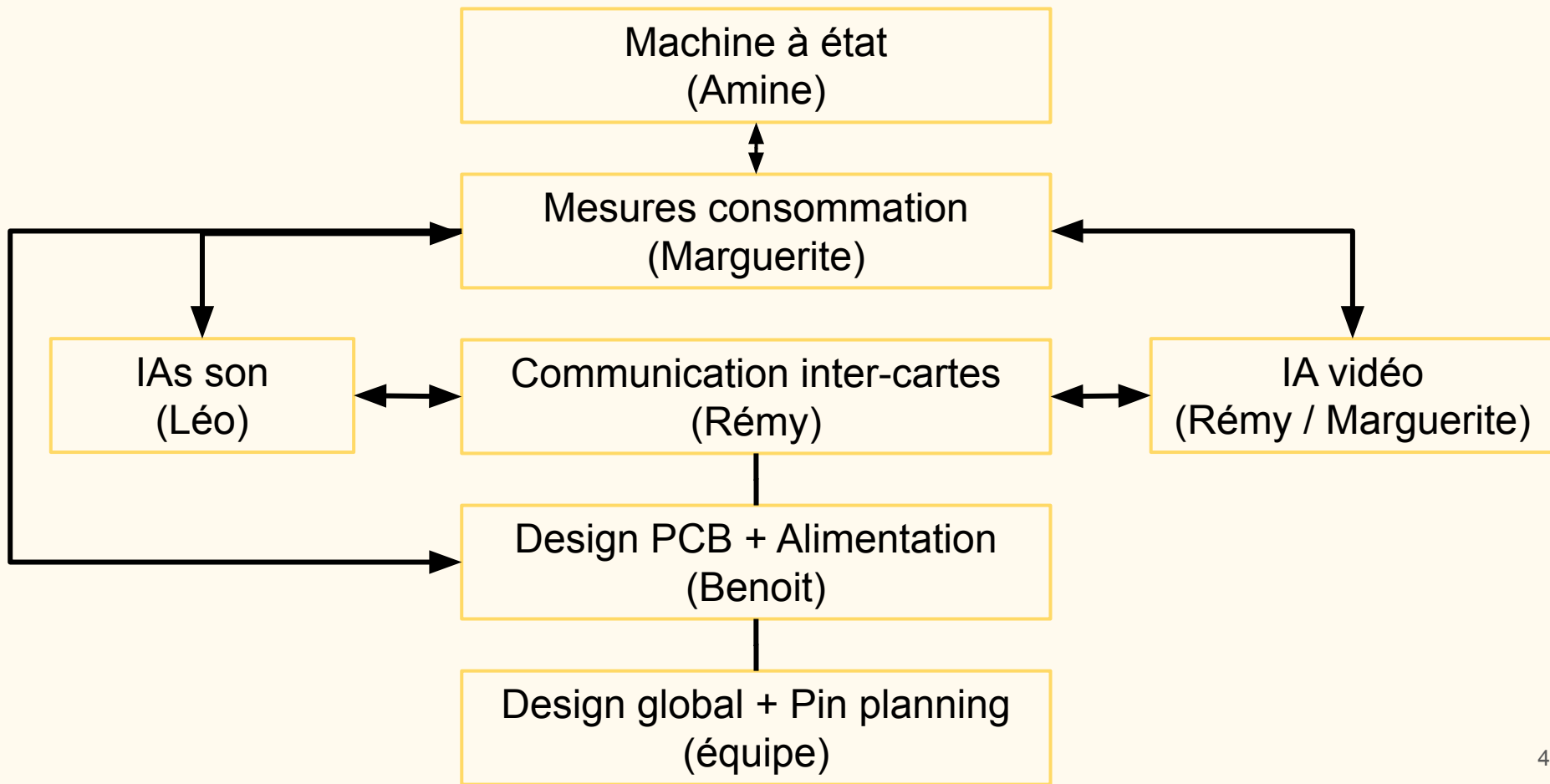


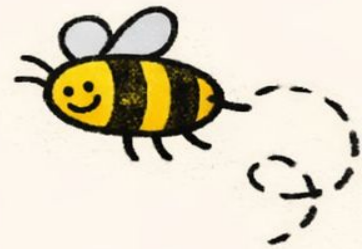
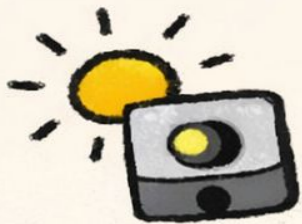
Contexte





Structuration idéale en équipe

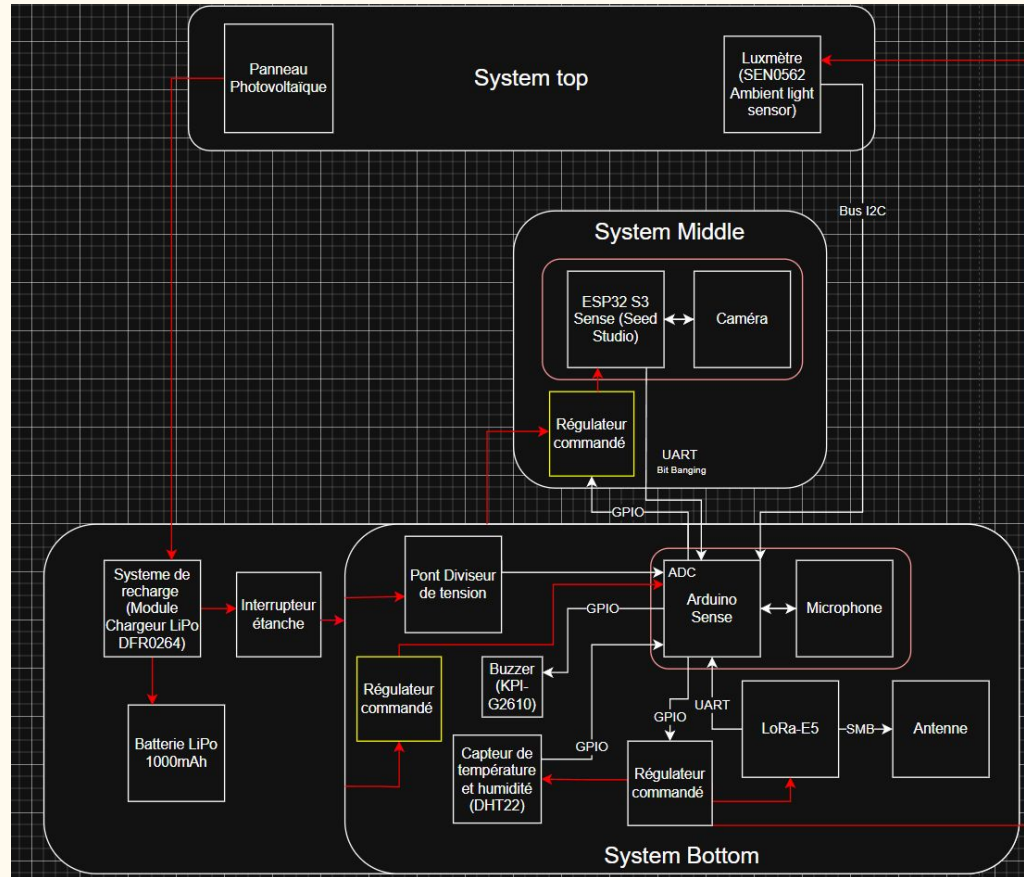




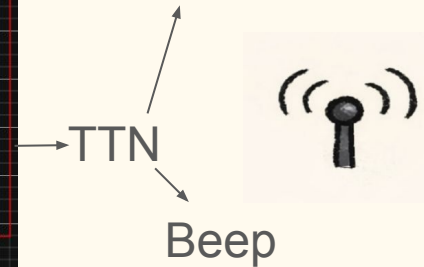
Structure finale



Schéma fonctionnel de la solution

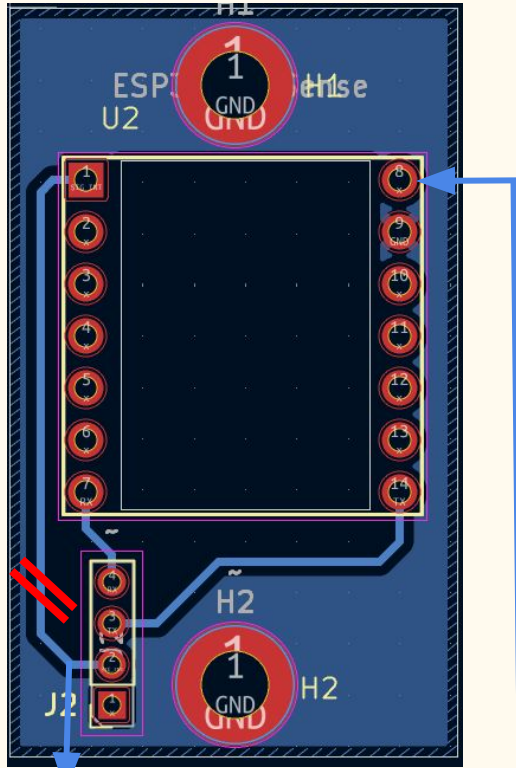


Pipedream
(alerte par mail)

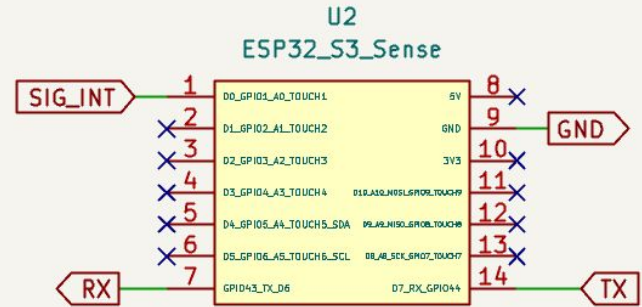
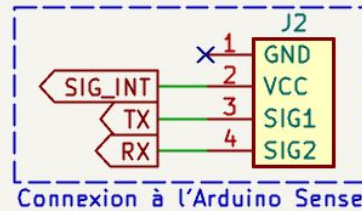




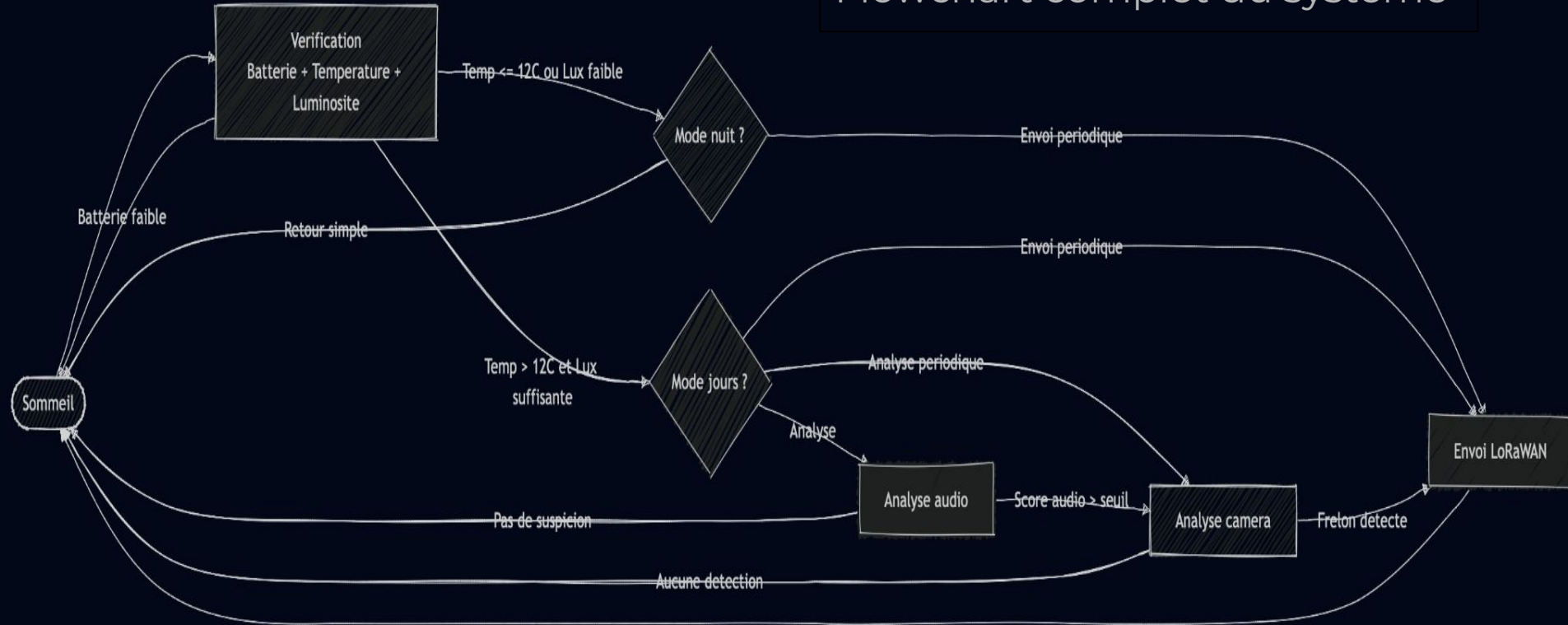
PCB & Schéma électronique XIAO

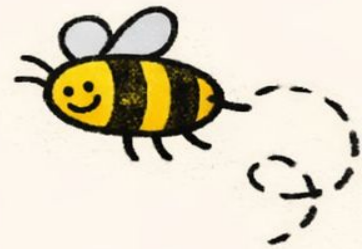
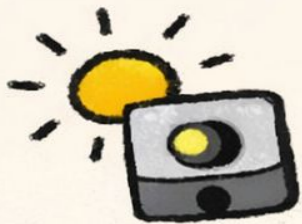


Régulateur
commandé 5V -> VIN

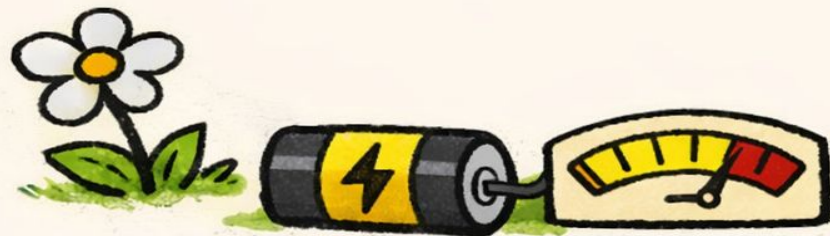


Flowchart complet du système

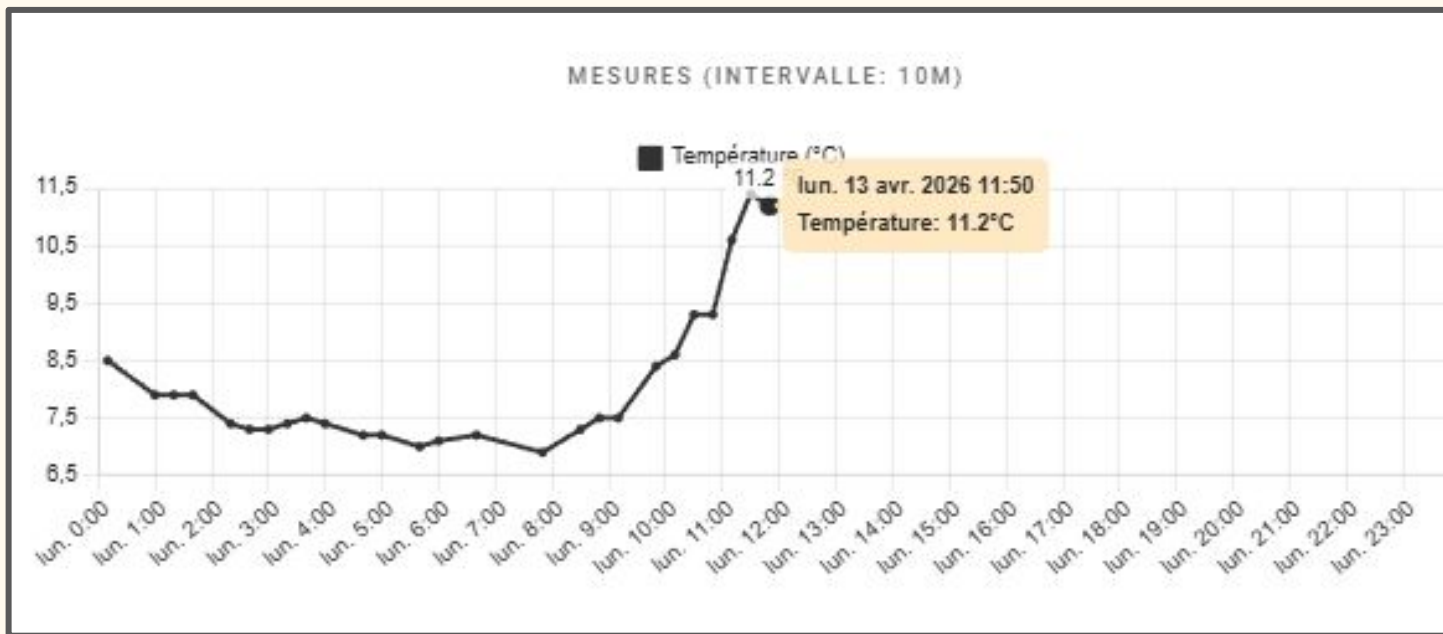
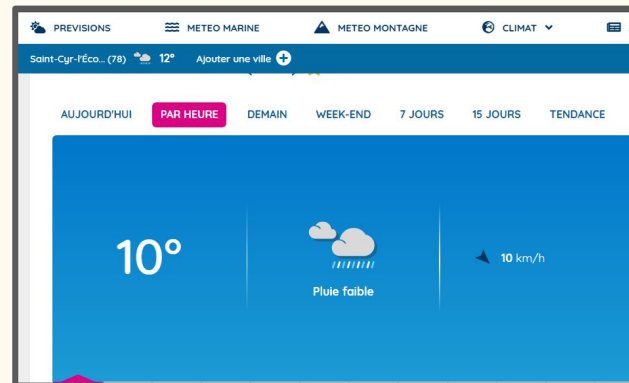




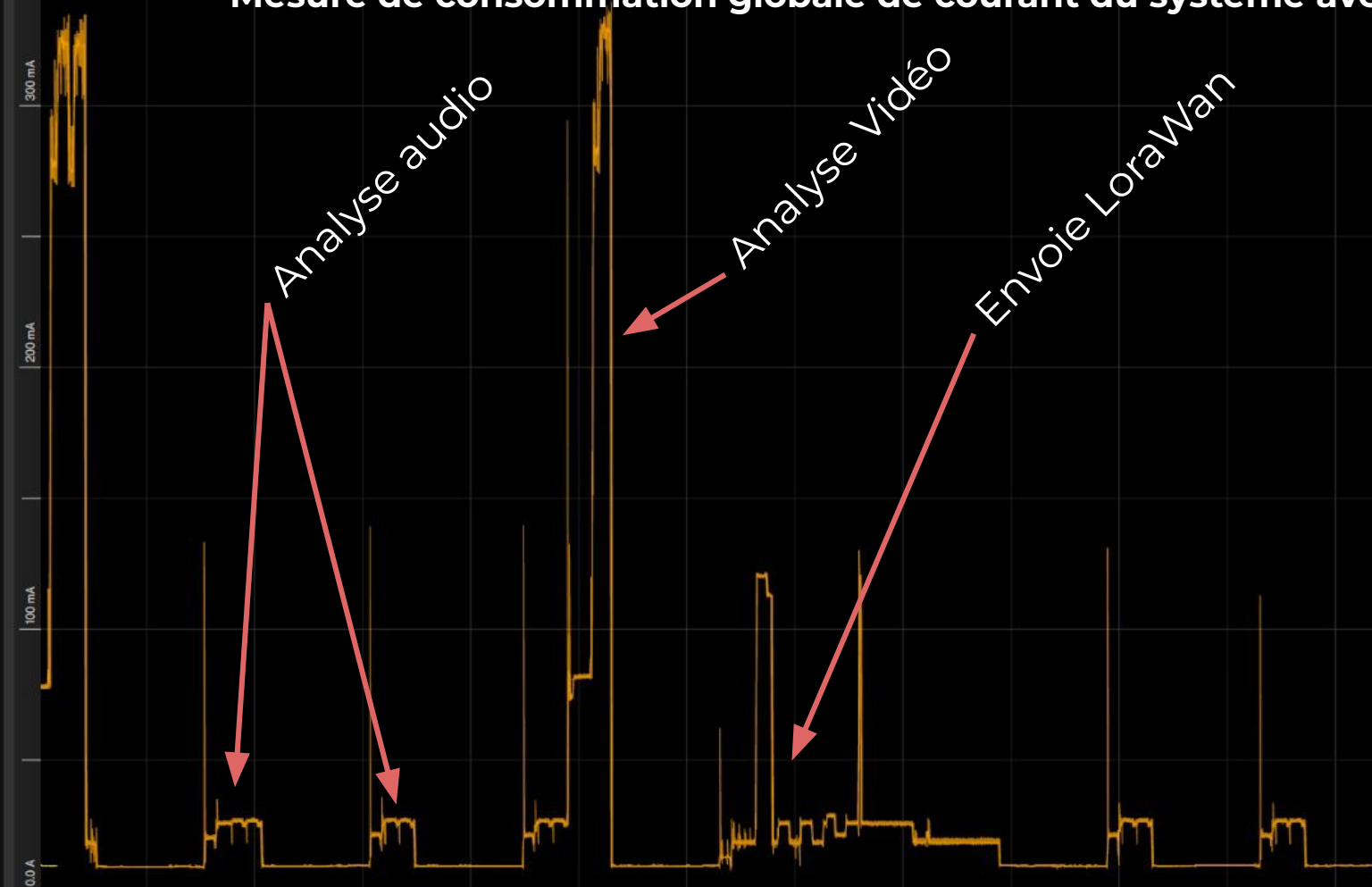
Mesures et consommation



Comparaison Mesure VS Météo

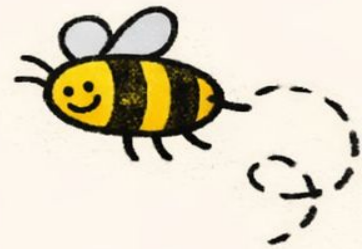
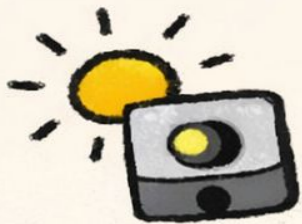


Mesure de consommation globale de courant du système avec Oti

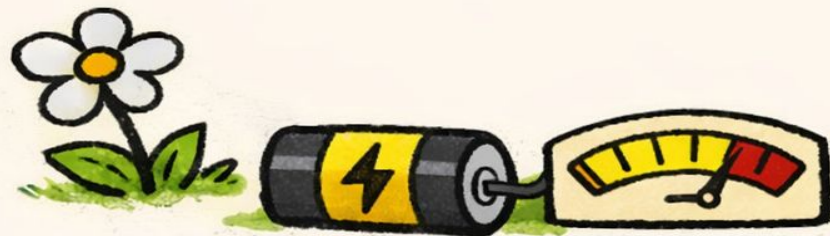


Max: 339 mA
Avg: 30.2 mA
Min: 8.30 mA
E: 3.46 mWh
C: 1.05 mAh

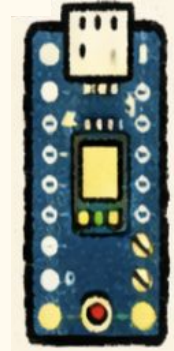
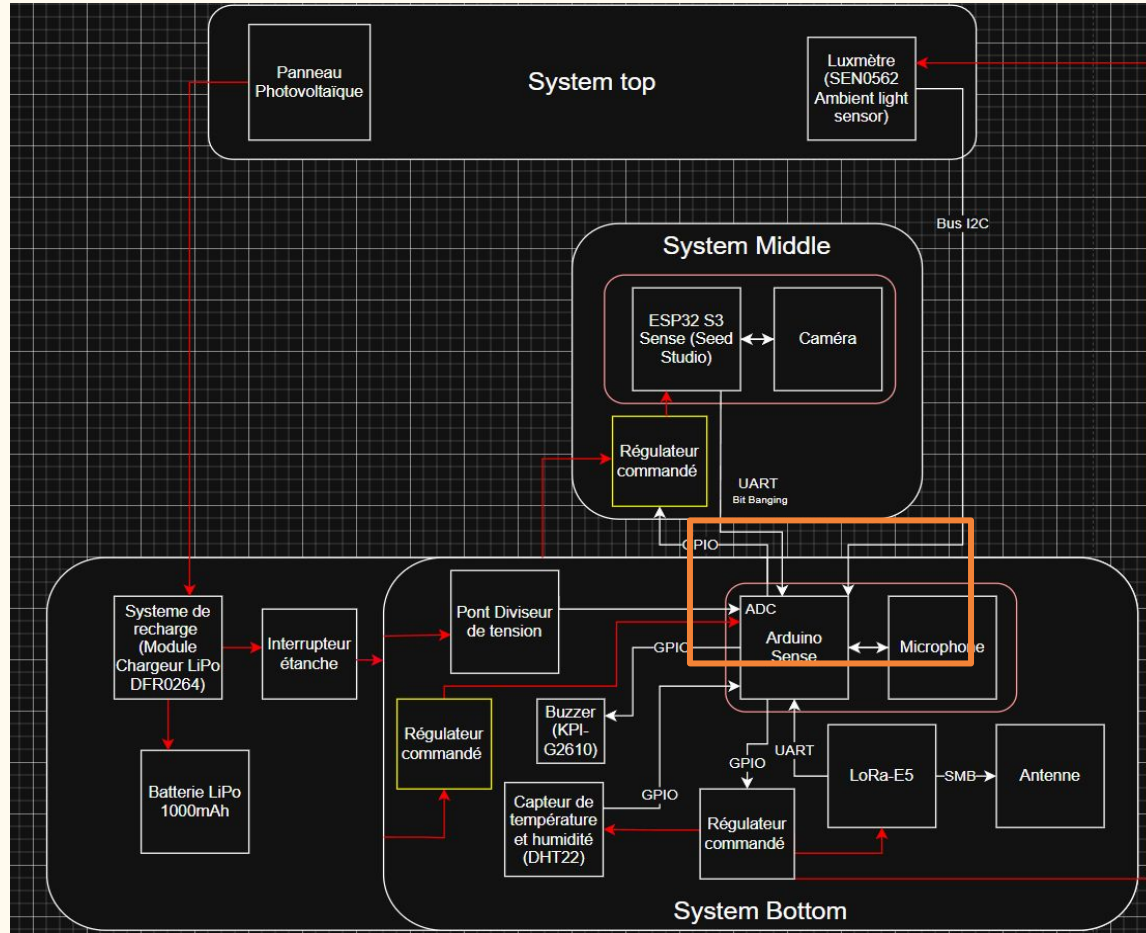
Max: 339 mA
Avg: 30.2 mA
Min: 8.30 mA
E: 3.46 mWh
C: 1.05 mAh



Ia Son & Image

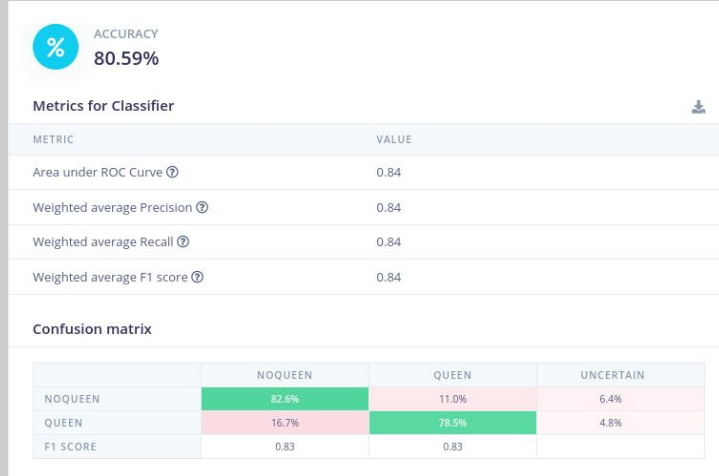


IA Son



Détection sonore des frelons et de la reine

Queen (MFE+1DConv) (Queen, No Queen)



Hornet (MFCC + 2DConv) (Hornet, No Hornet)

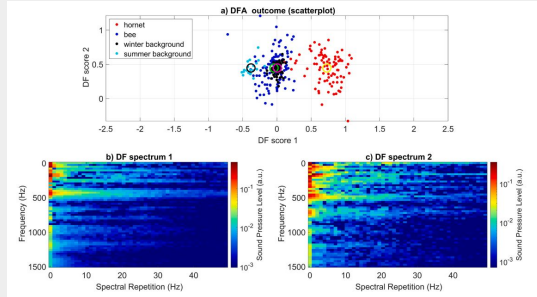


Multi-impulse transformation block Nécessite : requête API + modèles identiques (un seul fichier de configuration)

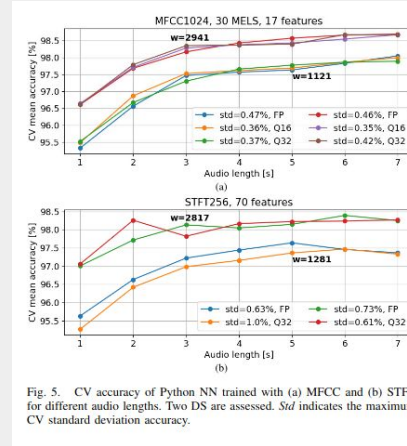
Modèles validés par tests sur Arduino Nano Sense avec dataset de test

Détection sonore des frelons et de la reine

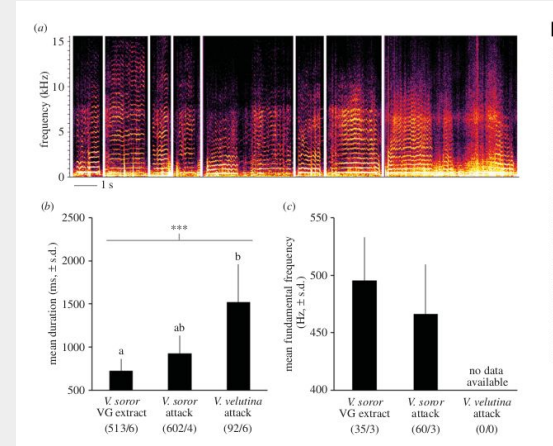
Solution actuelle



Détection du son émis par les frelons (1s de données)



Détection par activité de la ruche + sensor fusion
(mais 2-4 secondes de données)

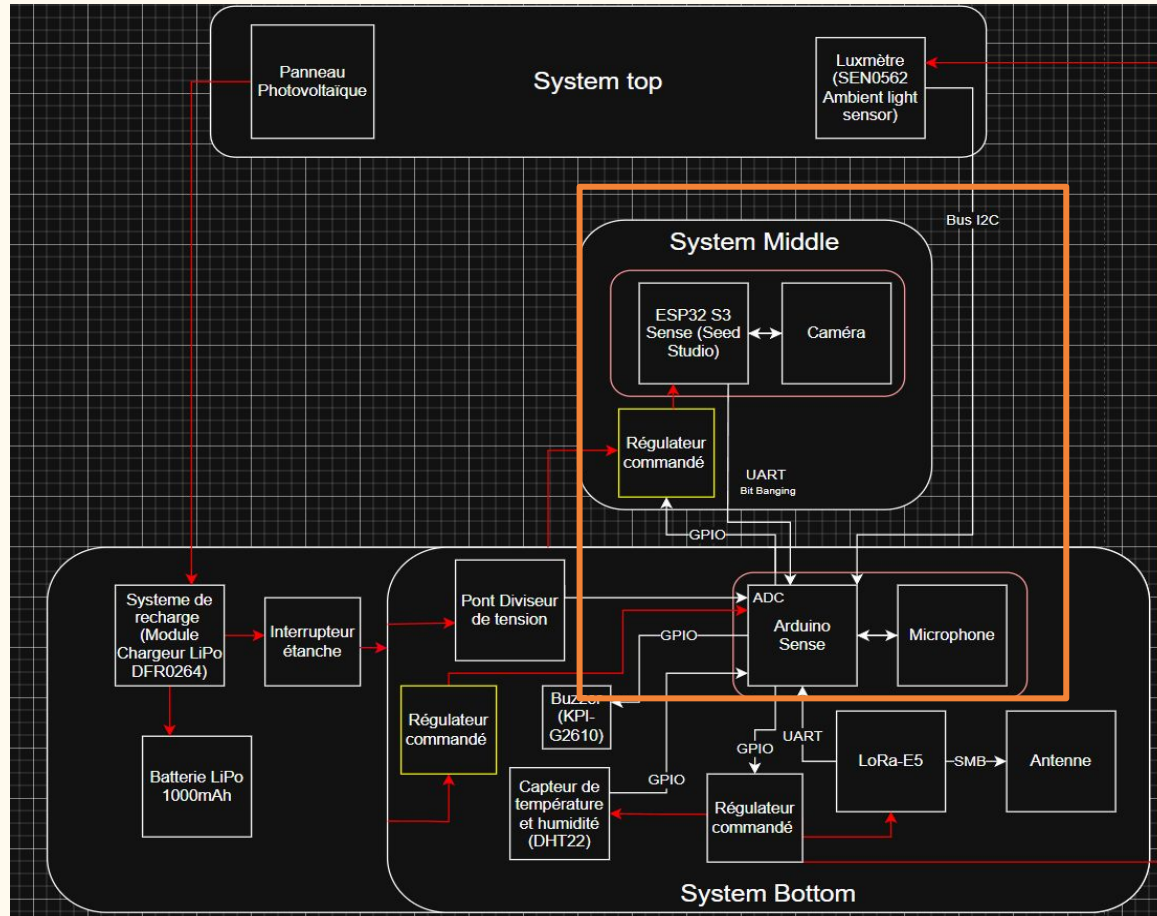


Meilleure précision, modèles compatibles **MAIS** plus gourmand en ressources

Espace mémoire occupé par le modèle de reconnaissance son

```
leo@summit-Precision-3581:~/Documents/Arduino$ arm-none-eabi-size -A /home/leo/.cache/arduino/sketches/D495481121573C8D1011B5741C4095BC/final.elf
section      size      addr
.text        260248      65536
.ARM.extab   292         325784
.ARM.exidx   496         326076
.data        4800      536871424
.nvictable   256      536870912
.crash_data_ram 256      536871168
.bss         76424      536876224
.heap        179384     536952648
.stack       1024      537132032
.ARM.attributes 46           0
.comment     247           0
.debug_info  2030621      0
.debug_abbrev 134057      0
.debug_loc   311437      0
.debug_aranges 10664      0
.debug_ranges 44432      0
.debug_macro  276327      0
.debug_line   447186      0
.debug_str    1619983      0
.debug_frame   52856      0
.stab         60           0
.stabstr      118           0
Total        5451214
```

IA Image



Last training performance (validation set)

%

F1 SCORE ?
82.1%

Confusion matrix (validation set)

	BACKGROUND	BEE	BEES	HORNET
BACKGROUND	100.0%	0.0%	0.0%	0.0%
BEE	14.3%	85.7%	0%	0%
BEES	17.1%	0%	82.9%	0%
HORNET	13.3%	0%	0%	86.7%
F1 SCORE	1.00	0.77	0.86	0.88

Metrics (validation set)

METRIC	VALUE
Precision (non-background) ?	0.79
Recall (non-background) ?	0.85
F1 Score (non-background) ?	0.82

On-device performance ?

Engine: ? TensorFlow Lite

INFERENCEING ...
6666 ms.

PEAK RAM USA...
1.6M

FLASH USAGE
110.5K

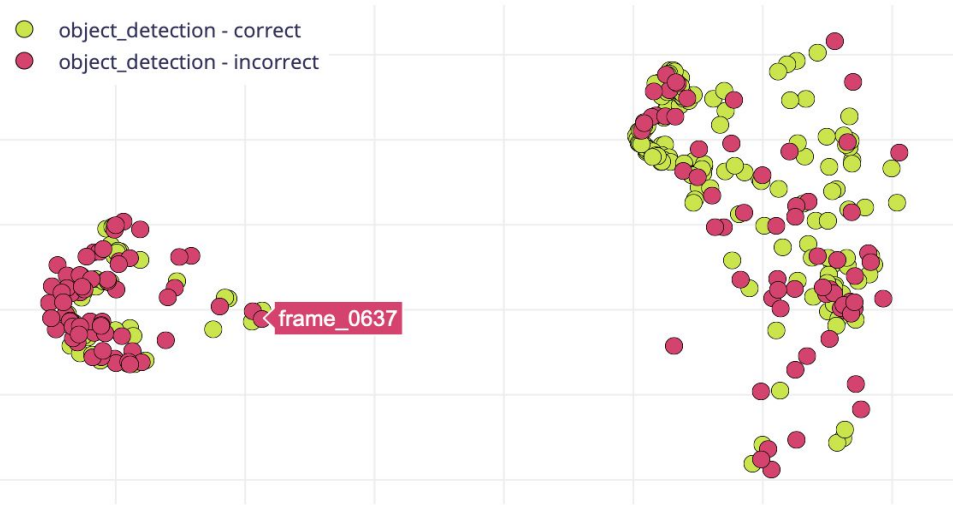
%

ACCURACY ?
61.54%

Metrics for Object detection

METRIC	VALUE
Precision (non-background) ?	0.77
Recall (non-background) ?	0.81
F1 Score (non-background) ?	0.79

Feature explorer ?





Résultats de l'IA avec Model 240x240



=====

===== DEBUG XIAO =====

[INFERENCE]

Bounding boxes count = 4

- Hornet | score=0.828 | percent=83% | x=56 y=24 w=16 h=8
- Bee | score=0.598 | percent=60% | x=8 y=88 w=8 h=8
- Bee | score=0.500 | percent=50% | x=144 y=128 w=8 h=8
- Bee | score=0.805 | percent=80% | x=152 y=152 w=16 h=16

[SCORES]

Hornet = 83%

Bee = 80%

Bees = 0%

[PAYLOAD]

136 83 80 0

=====

===== DEBUG XIAO =====

[INFERENCE]

Bounding boxes count = 7

- Bee | score=0.730 | percent=73% | x=176 y=104 w=16 h=8
- Bee | score=0.828 | percent=83% | x=200 y=112 w=8 h=8
- Bee | score=0.645 | percent=64% | x=112 y=136 w=8 h=8
- Bee | score=0.859 | percent=86% | x=224 y=144 w=8 h=8
- Bee | score=0.730 | percent=73% | x=112 y=168 w=8 h=8
- Bee | score=0.500 | percent=50% | x=168 y=176 w=8 h=8
- Bee | score=0.879 | percent=88% | x=192 y=184 w=8 h=8

[SCORES]

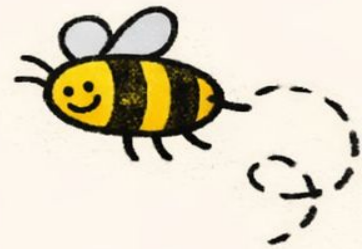
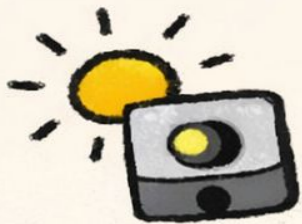
Hornet = 0%

Bee = 88%

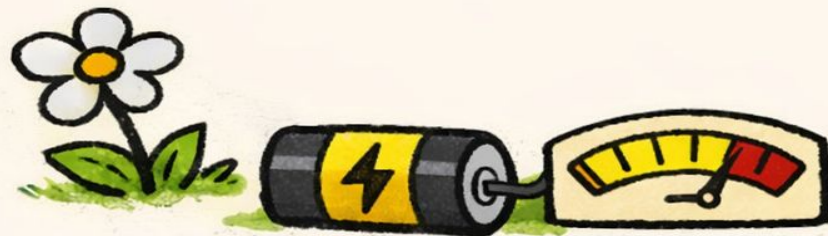
Bees = 0%

[PAYLOAD]

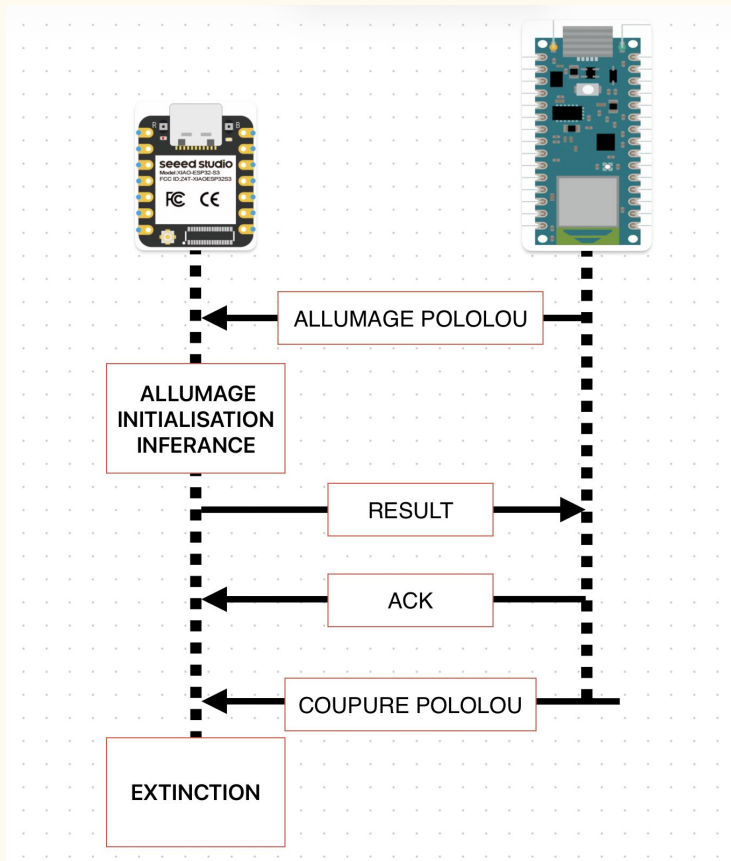
142 0 88 0



Communication inter-cartes

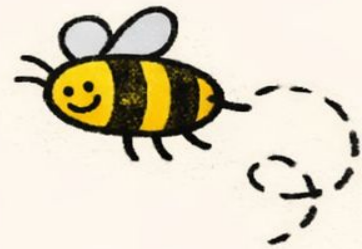
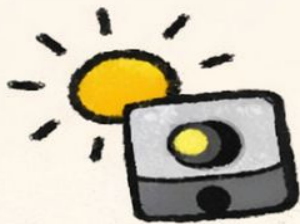


Flowchart communication entre les deux cartes



- **Chaque trame a cette structure :**
- [A5] [5A] [TYPE] [LEN] [PAYLOAD...] [CHK]
-
- **Types utilisés :**
- RESULT (TYPE = 0x02, LEN = 4)
- ERROR_MSG (TYPE = 0x05, LEN = 2)
-
- **Structure complète RESULT :**
- [A5] [5A] [02] [04] [frame_id] [hornet] [bee] [bees] [CHK]
-
- **ERROR_MSG :**
- [frame_id]
- [error_code]
-
- **Codes erreur :**
- 0x01 = erreur capture caméra
- 0x02 = erreur inférence/classifier





Resultats Beep / TTN



HEURE

JOUR

SEMAINE

MOIS

ANNÉE

SELECTION

Relative

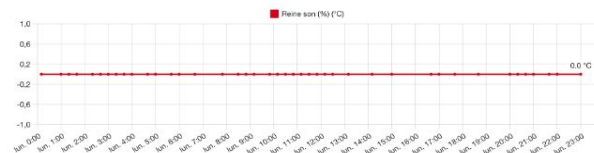
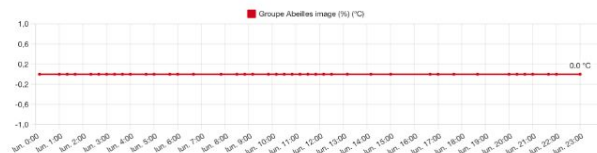
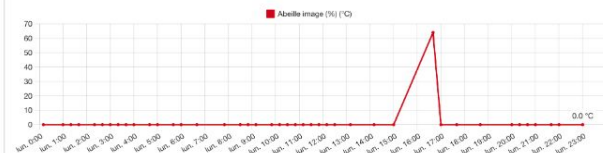
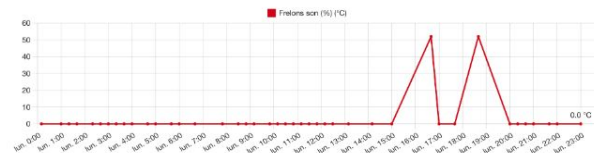
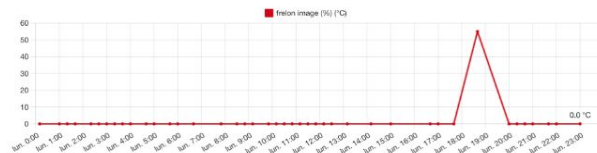
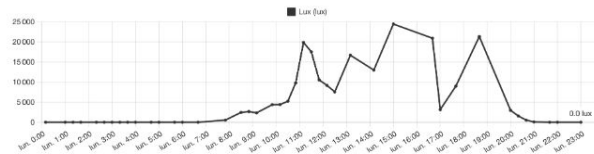
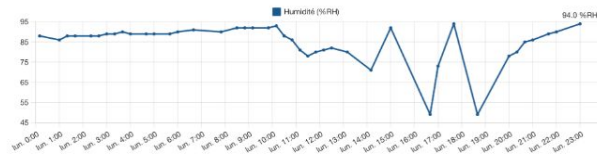
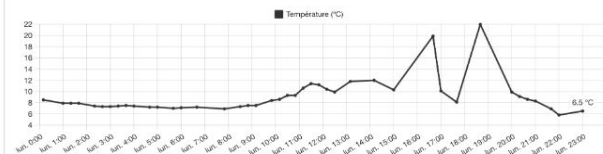
< LUN. 13 AVR. 2026 / >

Ruche Y1 - G1_FISA_2026_IAbelle

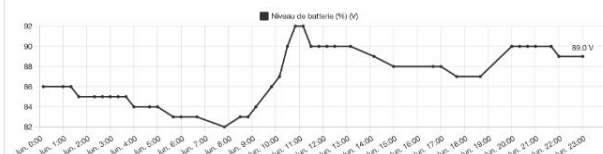


Mesures: Ruche Y1 - G1_FISA_2026_IAbelle

MESURES (INTERVALLE: 10M)



APPAREIL INFO




```

1 function decodeUplink(input) {
2   var b = input.bytes;
3   var data = {};
4
5   data.key = "76mqfbrqwa4qa6qb";
6
7   if (b && b.length >= 10) {
8     var tRaw = (b[0] << 8) | b[1]; // temperature
9     if (tRaw & 0x8000) tRaw -= 0x10000;
10    data.t = tRaw / 10.0;
11    data.t_1 = b[2]; // lastHornet
12    data.t_0 = b[3]; // lastStatus
13    data.l = (b[4] << 8) | b[5]; // lux
14    data.h = b[6];
15    data.bv = b[7];
16    data.t_2 = b[8]; //lastHornet_sound
17    data.rssi = -((b[9] << 8) | b[10]);
18    data.t_3 = b[11]; //lastBee
19    data.t_4 = b[12]; //lastBees
20    data.t_5 = b[13]; //lastQueen_sound
21  }
22
23  return {
24    data: data
25  };

```

Webhooks (2)

ID	BASE URL	TEMPLATE ID	STATUS
alerte	https://eob9htm7gwax26g.m.pipedream.net	None	Healthy •
iabeille	https://beep-test.azurewebsites.net/api/yann	None	Healthy •



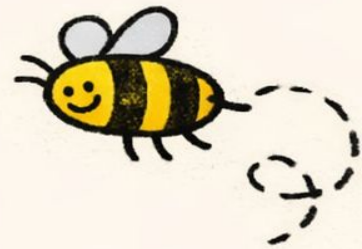
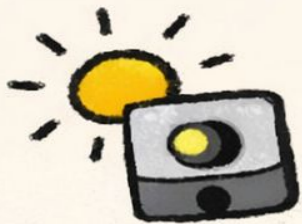
Alerte Frelon <iabeilles.su@gmail.com>

À moi ▾

- 📍 Informations de détection :- Nombre de frelons : 5
- Niveau d'alerte : CRITIQUE
- Timestamp : 2026-04-01T07:33:54.094Z

🔧 Informations des capteurs (Ruche Y6) :- Température : 25°C

- Humidité : 40%
- Batterie : 38V
- Luminosité : 600 Lux



Démonstration





Retours technique



Points limitants



Audio

- Concaténation de modèles difficile (bibliothèques C++)
- Arduino Nano Sense demandé par le client non adapté au projet (RAM limitée (enregistrement de 2s plus intéressant))
- Positionnement du microphone (VS Dataset)

Energie

- Gestion de l'énergie suffisante : estimation non précise mais validation empirique.

Machine à état

- MAE non intelligente : détection difficile.

IA

- **Manque de dataset** : Détection frelon par activité sonore de la ruche aurait été pertinente (nécessite 2-6 secondes)



Retour de gestion :

Methode Scrum / Post Scrum



Vers une organisation plus adaptée

Constats

- Les points courts et réguliers sont efficaces
- Les sprints aident à rythmer l'avancement
- La répartition initiale des tâches fonctionne bien

Axes d'apprentissage:

- Mieux cadrer dès le début du projet
- Prévoir un pilotage clair et identifié
- Renforcer les revues intermédiaires
- Mettre en place un suivi simple, en amont de projet
- Prévoir un échange client spécifique sur les données



Questions ?

